

每日一題46

單元2 多項式函數-降次

2025.10.16

114翰林第=次模考*15

已知 a 為 $x^2 - 4x + 1 = 0$ 的一個實數解，試求 $2a^4 - 9a^3 + 9a^2 - 12a - 3 + \frac{4}{a^2 + 1}$ 的值為

15-1 15-2

<Sol>

By 題意, $a^2 - 4a + 1 = 0 \dots (1)$

$$\textcircled{1} (1) \times 2a^2 \Rightarrow 2a^4 - 8a^3 + 2a^2 = 0 \Rightarrow 2a^4 - 8a^3 = -2a^2$$

$$\left[\begin{aligned} &2a^4 - 9a^3 + 9a^2 - 12a - 3 + \frac{4}{a^2 + 1} \\ &= -a^3 - 2a^2 + 9a^2 - 12a - 3 + \frac{4}{a^2 + 1} \\ &= -a^3 + 7a^2 - 12a - 3 + \frac{4}{a^2 + 1} \end{aligned} \right]$$

$$\textcircled{2} (1) \times (-a) \Rightarrow -a^3 + 4a^2 - a = 0 \Rightarrow -a^3 + 4a^2 = a$$

$$\left[\begin{aligned} &-a^3 + 7a^2 - 12a - 3 + \frac{4}{a^2 + 1} \\ &= 3a^2 + a - 12a - 3 + \frac{4}{a^2 + 1} = 3a^2 - 11a - 3 + \frac{4}{a^2 + 1} \end{aligned} \right]$$

$$\textcircled{3} (1) \times 3 \Rightarrow 3a^2 - 12a + 3 = 0 \Rightarrow 3a^2 - 12a = -3$$

$$\left[3a^2 - 11a - 3 + \frac{4}{a^2 + 1} = a - 3 - 3 + \frac{4}{a^2 + 1} = a + \frac{4}{a^2 + 1} - 6 \right]$$

$$\textcircled{4} \quad a^2 - 4a + 1 = 0 \Rightarrow a^2 + 1 = 4a$$

$$\left[a + \frac{4}{a^2+1} - b = a + \frac{4}{4a} - b = a + \frac{1}{a} - b \right]$$

$$\textcircled{5} \quad a^2 - 4a + 1 = 0 \Rightarrow a - 4 + \frac{1}{a} = 0 \Rightarrow a + \frac{1}{a} = 4$$

$$\left[a + \frac{1}{a} - b = 4 - b = -2 \right] \#$$